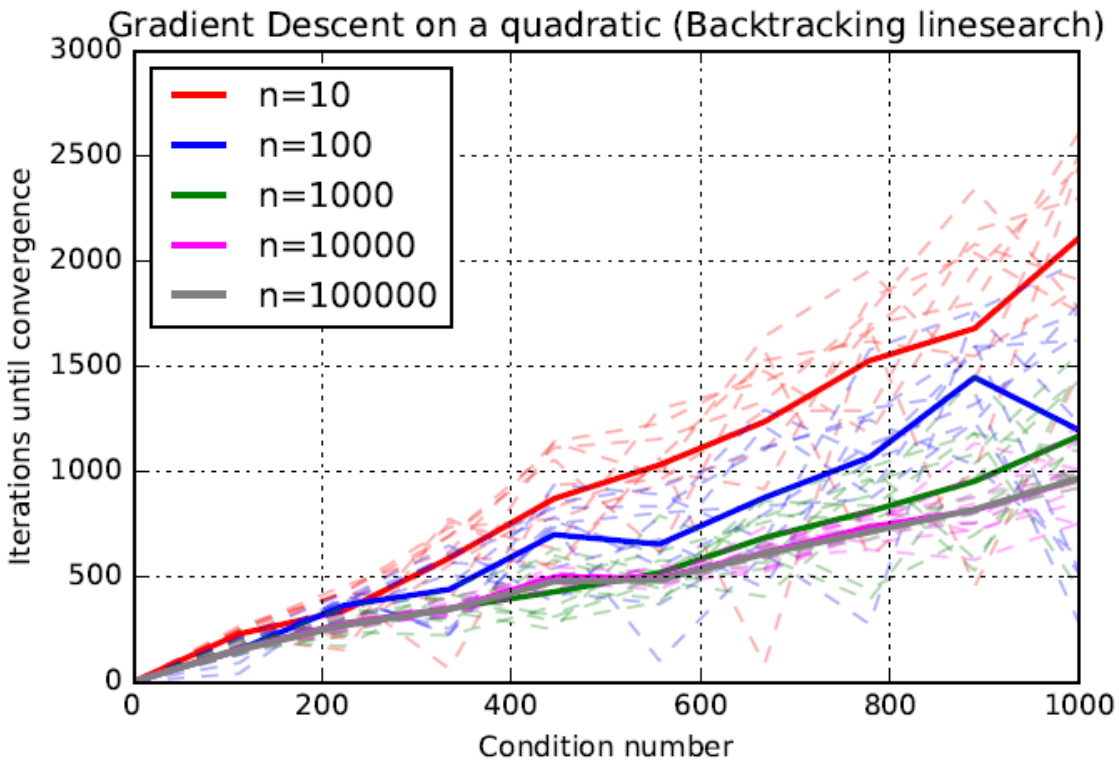
Non-linear Conjugate Gradient Method

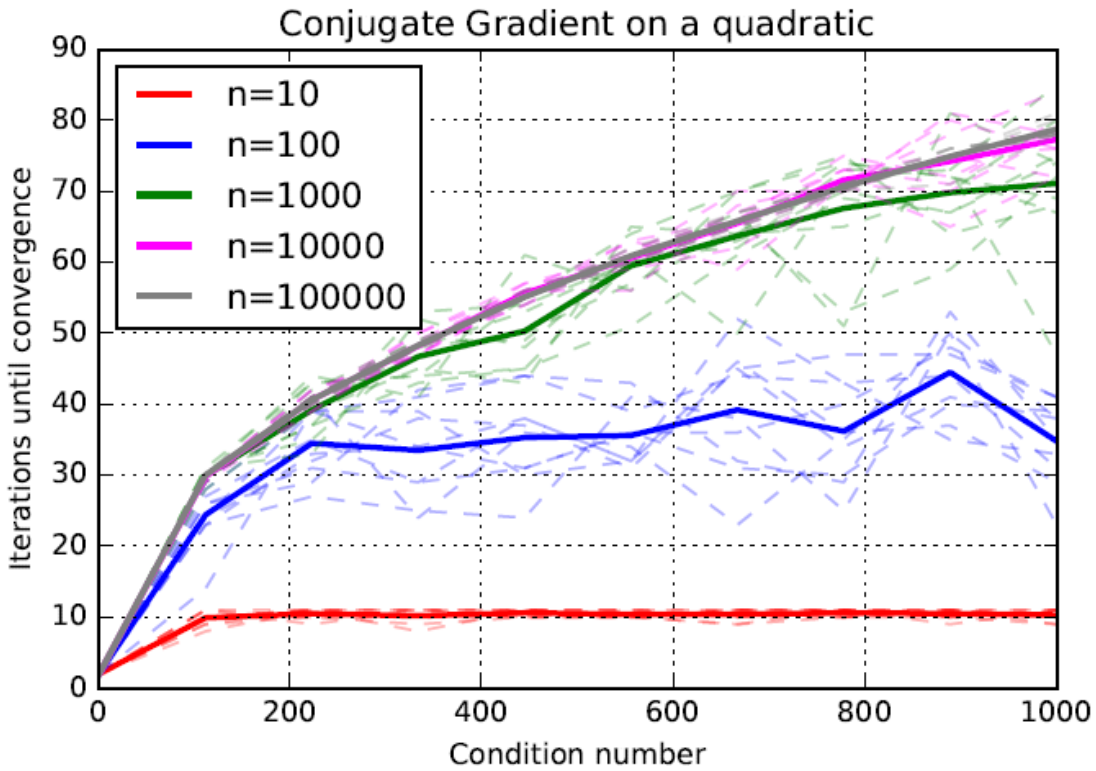
Linear CG

- Init $\boldsymbol{g}_0 = A\boldsymbol{x}_0 - \boldsymbol{b}$, $\boldsymbol{d}_0 = -\boldsymbol{g}_0$, $\boldsymbol{u}_0 = A\boldsymbol{d}_0$.
- For $k = 0, 1, 2, \ldots$:
  - $\alpha_k = \frac{\boldsymbol{g}_k^T \boldsymbol{g}_k}{\boldsymbol{d}_k^T \boldsymbol{u}_k}$;
  - $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{d}_k$; $\boldsymbol{g}_{k+1} = \boldsymbol{g}_k + \alpha_k \boldsymbol{u}_k$;
  - If $\|\boldsymbol{g}_{k+1}\|^2 \leq \varepsilon$, then stop.
  - $\beta_k = \frac{\boldsymbol{g}_{k+1}^T \boldsymbol{g}_{k+1}}{\boldsymbol{g}_k^T \boldsymbol{g}_k}$; $\boldsymbol{d}_{k+1} = -\boldsymbol{g}_{k+1} + \beta_k \boldsymbol{d}_k$;
  - $\boldsymbol{u}_{k+1} = A\boldsymbol{d}_{k+1}$.

Linear CG performance

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T A\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b} \to \min_{\boldsymbol{x}} \quad A = \mathrm{diag}(a_1, \ldots, a_n);\ \boldsymbol{b} \sim \mathcal{N}(0, I);\ a_i = \begin{cases} 1,\ i = 1, \\ \sim \mathrm{Unif}(1, \kappa),\ 2 \le i \le n-1; \\ \kappa,\ i = n. \end{cases}$$





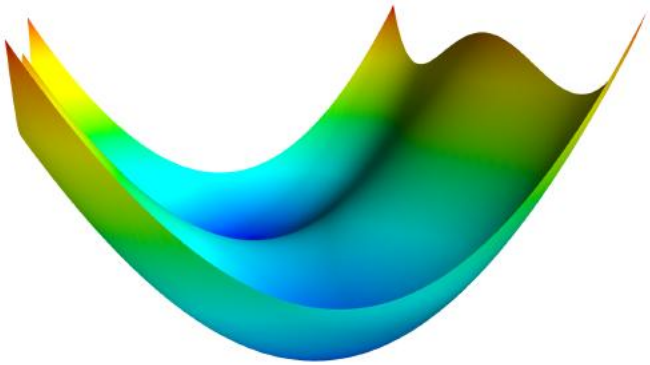
Fletcher-Reeves method

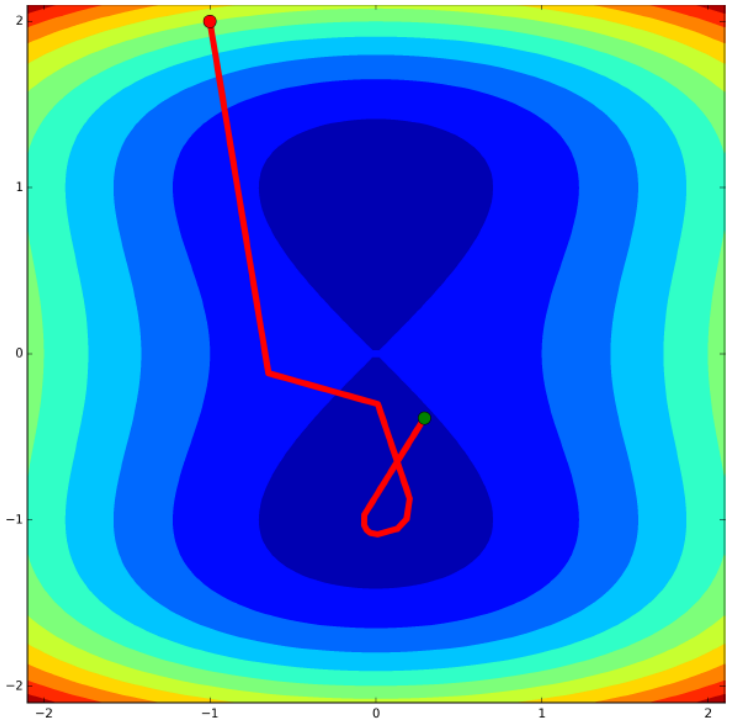
Idea: take linear CG scheme and change the corresponding parts for arbitrary function

- Init $\boldsymbol{g}_0 = \nabla f(\boldsymbol{x}_0)$, $\boldsymbol{d}_0 = -\boldsymbol{g}_0$.
- For $k = 0, 1, 2, \ldots$:
  - α_k is determined by line search using Armijo/Wolfe;
  - $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{d}_k$;
  - $\boldsymbol{g}_{k+1} = \nabla f(\boldsymbol{x}_{k+1})$;
  - If $\|\boldsymbol{g}_{k+1}\|^2 \leq \varepsilon$, then stop.
  - $\beta_k = \frac{\boldsymbol{g}_{k+1}^T \boldsymbol{g}_{k+1}}{\boldsymbol{g}_k^T \boldsymbol{g}_k}$; $\boldsymbol{d}_{k+1} = -\boldsymbol{g}_{k+1} + \beta_k \boldsymbol{d}_k$.

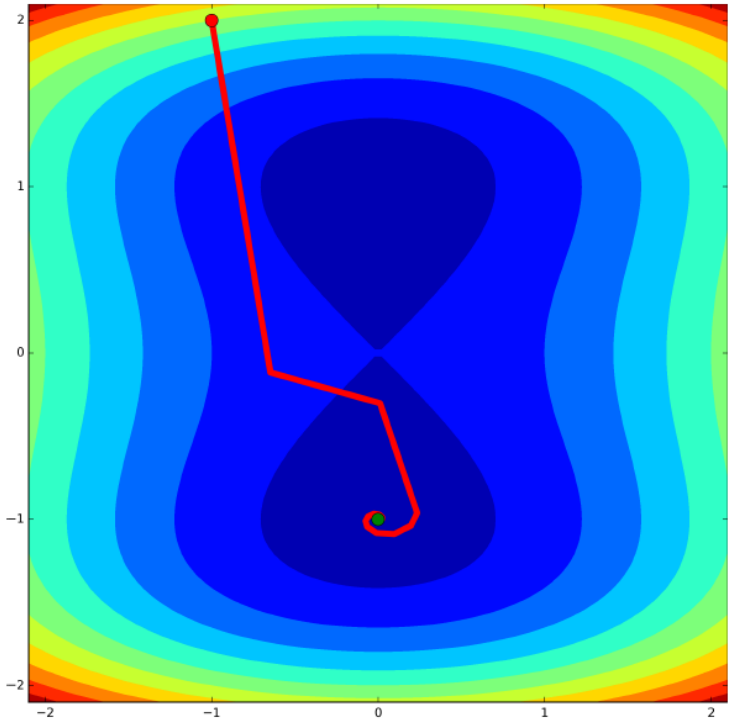
Fletcher-Reeves and descent direction

Example: $f(x,y) = \frac{1}{2}x^2 + \frac{1}{4}y^4 - \frac{1}{2}y^2 \to \min_{x,y}$





$c_2 = 0.9$



$c_2 = 0.2$

Descent direction in FR method: use Armijo/Wolfe with c2 < 0.5

Alternative non-linear CG methods

Use FR scheme with different formula for beta:

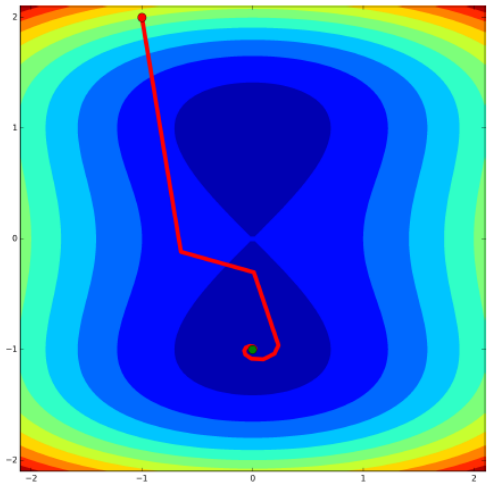
- Polak-Ribiere: $\beta_k^{PR} = \frac{\boldsymbol{g}_{k+1}^T(\boldsymbol{g}_{k+1}-\boldsymbol{g}_k)}{\|\boldsymbol{g}_k\|^2}$;
- Hestenes-Stiefel: $\beta_k^{HS} = \frac{\boldsymbol{g}_{k+1}^T(\boldsymbol{g}_{k+1}-\boldsymbol{g}_k)}{\boldsymbol{d}_k^T(\boldsymbol{g}_{k+1}-\boldsymbol{g}_k)}$;
- Polak-Ribiere+: $\beta_k = \max(0, \beta_k^{PR})$;
- Gilbert-Nocedal: $\beta_k^{GN} = \max(-\beta_k^{FR}, \min(\beta_k^{PR}, \beta_k^{FR}))$.

Alternative non-linear CG methods

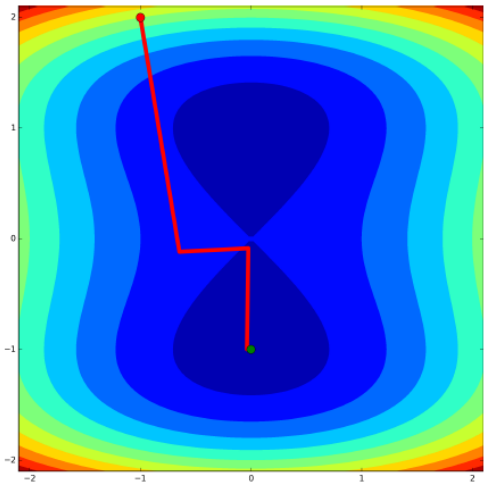
Use the idea of restarts:

If $|\boldsymbol{g}_{k+1}^T \boldsymbol{g}_k| > \nu \|\boldsymbol{g}_{k+1}\| \|\boldsymbol{g}_k\|$, then $\beta_{k+1} = 0$ $(\boldsymbol{d}_{k+1} = -\boldsymbol{g}_{k+1})$. $\nu = 0.1$

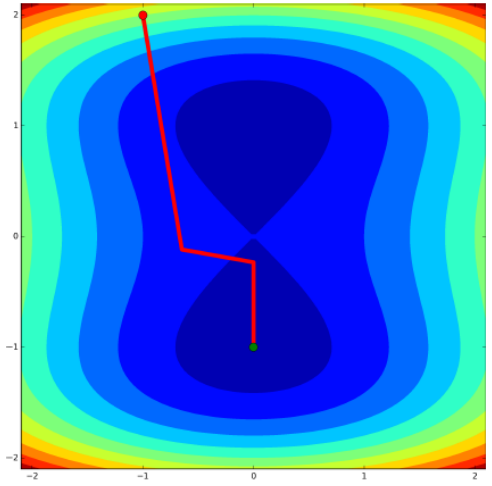
FR vs PR vs PR+restart



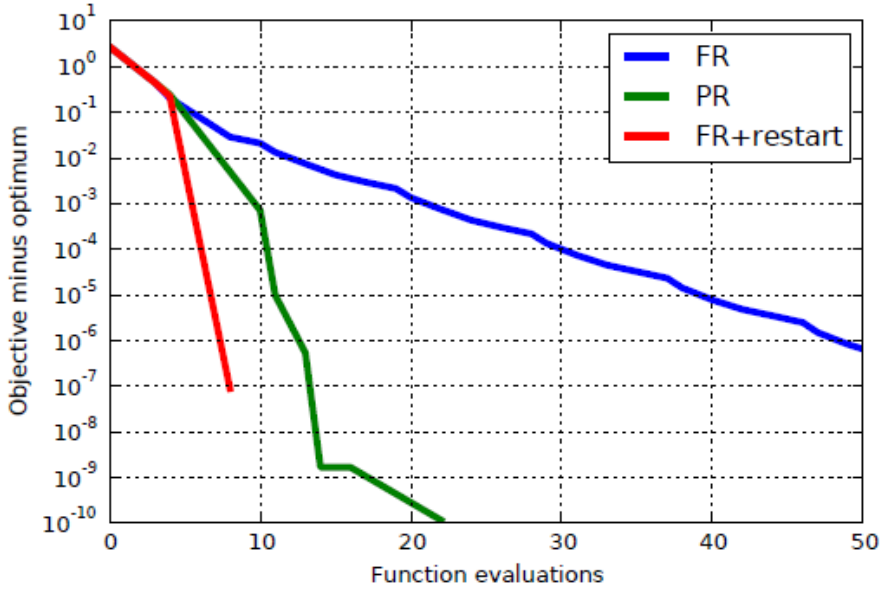
FR



PR

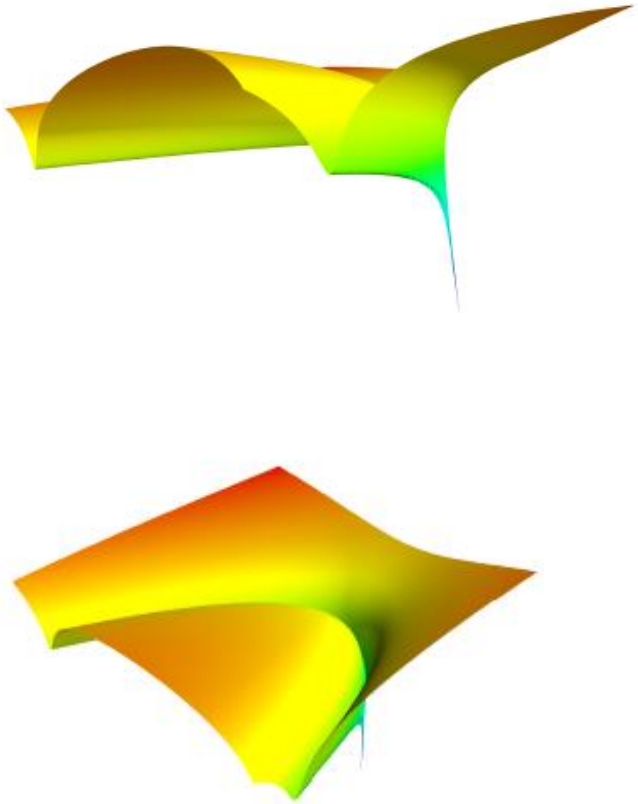


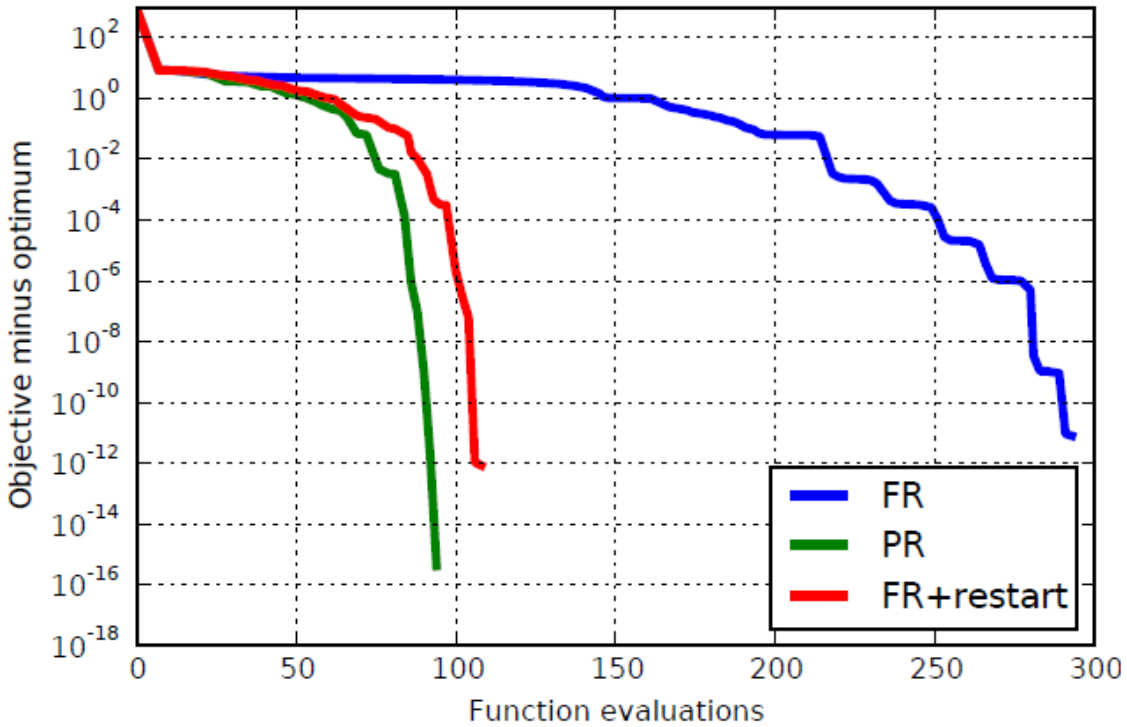
FR+restart



Example: Rosenbrock function

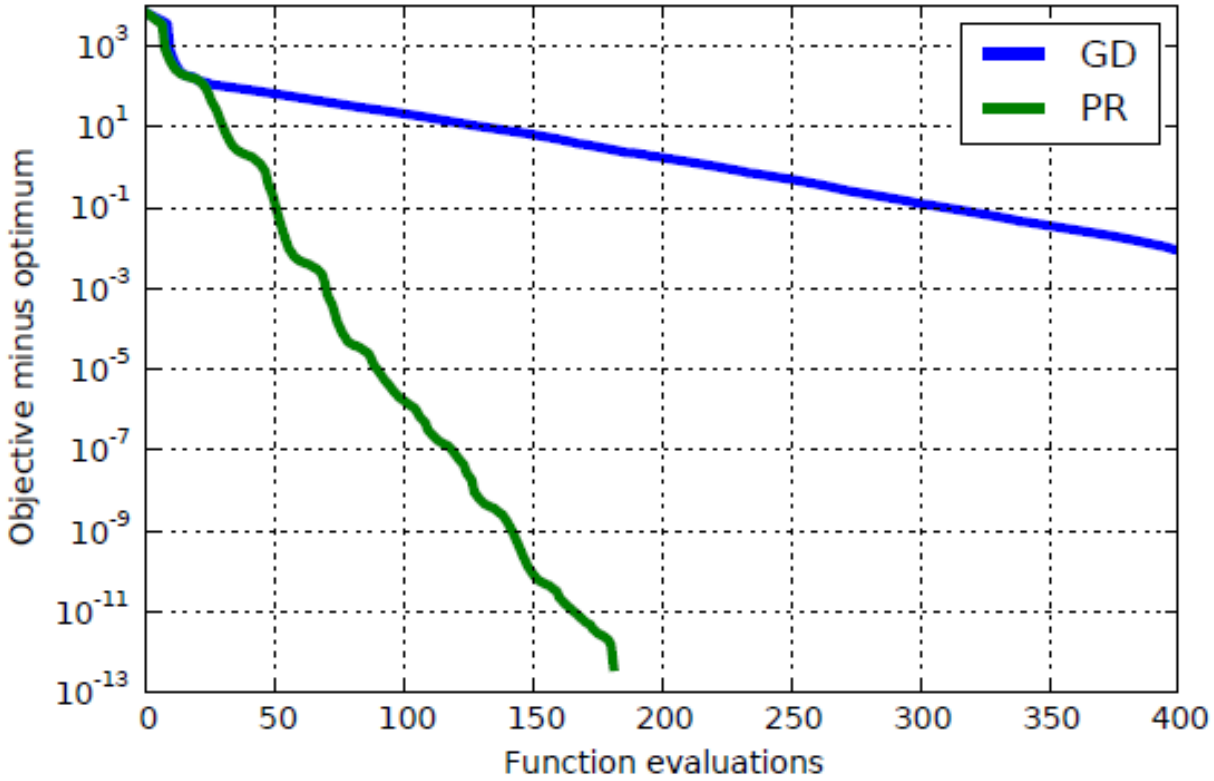
$$f(x,y) = (1-x)^2 + 100(y-x^2)^2 \to \min_{x,y}$$





Example: logistic regression

$$f(\boldsymbol{w}) = \frac{1}{N}\sum_{i=1}^{N}\log(1+\exp(-y_i\boldsymbol{w}^T\boldsymbol{x}_i) + \frac{\lambda}{2}\|\boldsymbol{w}\|^2 \to \min_{\boldsymbol{w}}$$



CG vs GD

GD: Memory $O(n)$, Iteration cost $O(n)$, linear conv. rate with $C = 1 - \frac{\mu}{L}$

CG: Memory $O(n)$, Iteration cost $O(n)$, linear conv. rate with $C \approx 1 - \sqrt{\frac{\mu}{L}}$